

Regressions for Final 590 Project

2026-04-28

R Markdown

Libraries I need

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.2.1    ✓ readr      2.2.0
## ✓ forcats    1.0.1    ✓ stringr    1.6.0
## ✓ ggplot2     4.0.3    ✓ tibble     3.3.1
## ✓ lubridate  1.9.5    ✓ tidyr      1.3.2
## ✓ purrr      1.2.2
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(WDI)
library(stargazer)
```

```
##
## Please cite as:
##
## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

Importing Data

```
data1 <- read_csv("C:/Users/lamas/OneDrive/Documents/eco590/final-project-Jesslamastra/mlb_data.csv")
```

```
## Rows: 429 Columns: 8
## — Column specification —
## Delimiter: ","
## chr (3): Player, Position, Team
## dbl (5): OPS, HR, BattingAvg, RBI, Salary
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
data2 <- read_csv("C:/Users/lamas/OneDrive/Documents/eco590/final-project-Jesslamastra/mlb_data
2.csv")
```

```
## Rows: 115 Columns: 10
## — Column specification —————
## Delimiter: ","
## chr (3): Player, Position, Team
## dbl (7): OPS, HR, BattingAvg, RBI, Salary, OFF, WAR
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Cleaning Data

1. Drop all players in data set where batting average = 0 (mostly pitchers and players that were hurt all season. I drop this because it would skew my results in a negative way) I only drop these values from data1 because pitchers still have WAR and OFF stats that I don't want dropped
2. Encode position to dummy variable so I can include in regression
3. Encode team to dummy variable so it can be included in regressions

```
data1 <- data1[data1$OPS != 0 & data1$OPS != 2, ] #gets rid of rows where ops = 0 and 2

data1$Position <- relevel(factor(data1$Position), ref = "OF") #changes position so i can use it
in regressions
data2$Position <- relevel(factor(data2$Position), ref = "OF")

data1$Team <- relevel(factor(data1$Team), ref = "Yankees") #changes team so i can use it in regr
essions
data2$Team <- relevel(factor(data2$Team), ref = "Yankees")
```

Regressions

Regression using 3 variables Position omitted category is outfielder

```
model <- lm(Salary ~ OPS + BattingAvg + HR + RBI + Position, data = data1)
summary(model)
```

```
##
## Call:
## lm(formula = Salary ~ OPS + BattingAvg + HR + RBI + Position,
##     data = data1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16233697 -4307784 -1205025  2118162 45982189
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1785176    2908830  -0.614   0.5398
## OPS             13046425   10231081   1.275   0.2031
## BattingAvg    -21214087   24538433  -0.865   0.3879
## HR              188106     130682    1.439   0.1509
## RBI              30028      40947     0.733   0.4638
## Position1B     252601     1480166   0.171   0.8646
## Position2B    -460489     1436455  -0.321   0.7487
## Position3B     1569316    1500978   1.046   0.2965
## PositionC     -1482498    1258587  -1.178   0.2396
## PositionDH     2215210    1994098   1.111   0.2674
## PositionDH/RHP  9343271    8094135   1.154   0.2491
## PositionINF    -2200902    1673832  -1.315   0.1894
## PositionRHP    -1329251    7699578  -0.173   0.8630
## PositionSS     2622197    1426903   1.838   0.0669 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7653000 on 357 degrees of freedom
## Multiple R-squared:  0.2431, Adjusted R-squared:  0.2155
## F-statistic: 8.82 on 13 and 357 DF, p-value: 1.101e-15
```

Results interpretation

My regression shows that for a baseball player that has 0 ops, 0 batting average, and play no position a player is expected to make -8,113,324. This is just a projection on what players make, I have no instances in the data set that satisfy these circumstances which is why the number is a bit non-nonsensical.

I quickly notice my regression doesn't make sense because of the scale of my variables. since batting average is always less and one and ops typically doesn't exceed 1.5 the interpretations are incorrect.

To fix this I am going to log Salary.

```
model2 <- lm(log(Salary) ~ OPS + BattingAvg + HR + RBI + Position, data = data1)
summary(model2)
```

```
##
## Call:
## lm(formula = log(Salary) ~ OPS + BattingAvg + HR + RBI + Position,
##     data = data1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.2863 -0.9061 -0.1416  0.9277  2.8338
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   13.61196    0.44685   30.462  <2e-16 ***
## OPS           2.33683    1.57168    1.487   0.1379
## BattingAvg    -4.52492    3.76954   -1.200   0.2308
## HR            -0.01007    0.02007   -0.501   0.6164
## RBI           0.01503    0.00629    2.389   0.0174 *
## Position1B    -0.01763    0.22738   -0.078   0.9382
## Position2B    -0.01152    0.22066   -0.052   0.9584
## Position3B     0.23854    0.23058    1.035   0.3016
## PositionC     -0.08557    0.19334   -0.443   0.6583
## PositionDH     0.41263    0.30633    1.347   0.1788
## PositionDH/RHP 1.47083    1.24340    1.183   0.2376
## PositionINF   -0.53476    0.25713   -2.080   0.0383 *
## PositionRHP   -0.58857    1.18279   -0.498   0.6191
## PositionSS     0.19253    0.21920    0.878   0.3804
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.176 on 357 degrees of freedom
## Multiple R-squared:  0.1919, Adjusted R-squared:  0.1625
## F-statistic: 6.522 on 13 and 357 DF,  p-value: 3.668e-11
```

After logging salary

A .1 increase in on base plus slugging (ops) results in a 23% increase in salary.

1 additional rbi(runs batted in) is associated with a 1.5% increase in yearly salary.

This makes more sense than my first regression. However the coefficient on batting average and home runs is still negative suggesting an increase in batting average and home runs results in a decrease of salary which doesn't make sense economically. Since OPS (on base plus slugging percentage) is derived from components that overlap with batting average and home runs, the variables are highly correlated. This makes it difficult to isolate their individual effects on salary.

Additionally, the only statistically significant variable in this regression is RBI and Position Infield. This is because of the multicollinearity that is observed however they are still important variables to include.

I am going to drop OPS statistic to see the difference in coefficients.

Question answered: What Batting Metric affects player salary most?

```
model3 <- lm(log(Salary) ~ HR + BattingAvg + RBI + Position, data = data1)
summary(model3)
```

```
##
## Call:
## lm(formula = log(Salary) ~ HR + BattingAvg + RBI + Position,
##     data = data1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4861 -0.8955 -0.1433  0.9365  2.6581
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  13.952788   0.384219  36.315  <2e-16 ***
## HR           0.008712   0.015631   0.557   0.5776
## BattingAvg   0.376592   1.831173   0.206   0.8372
## RBI          0.012363   0.006040   2.047   0.0414 *
## Position1B   0.009223   0.227044   0.041   0.9676
## Position2B  -0.033312   0.220550  -0.151   0.8800
## Position3B   0.228410   0.230866   0.989   0.3232
## PositionC   -0.067045   0.193265  -0.347   0.7289
## PositionDH   0.430126   0.306620   1.403   0.1615
## PositionDH/RHP 1.356191   1.243108   1.091   0.2760
## PositionINF  -0.561566   0.256931  -2.186   0.0295 *
## PositionRHP  -0.577922   1.184770  -0.488   0.6260
## PositionSS   0.150895   0.217769   0.693   0.4888
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.178 on 358 degrees of freedom
## Multiple R-squared:  0.1869, Adjusted R-squared:  0.1597
## F-statistic: 6.858 on 12 and 358 DF,  p-value: 3.536e-11
```

Results

The regression logging salary and dropping OPS makes a lot of sense and can now be interpreted.

My regression shows if you have 0 home runs, a 0 percent batting average you will expect to get paid around 14 dollars per year. This instance is not in my data set.

1 additional home run leads to a .87% increase of yearly salary holding batting average, RBI, and position constant.

A 0.1 increase in batting average leads to a 3.8% increase in yearly salary for MLB players holding home runs, RBI, and position constant

One additional RBI leads to a 1.23% increase in yearly salary holding batting average, home runs, and position constant

For position the omitted category is outfielder. So holding batting average, rbi, and home runs constant: - 1 baseman make .92% more per year than outfielders

- 2 baseman makes 3.33% less than outfielders per year
- 3 baseman make 22.84% more per year than outfielders
- Catchers make 6.7% less than outfielders per year

- Designated Hitters make 43.01% more than outfielders per year
- DH and RHP (only 1 player in league pitches and hits - Ohtani) and makes 135.61% more per year than outfielders
- Utility infielders make 56.16% less per year than outfielders
- Right handed pitchers make 57.79% less per year than outfielders
- Short Stops make 15.09% more than outfielder per year

I do want to mention in this regression that only RBI and position infield are statistically significant to salary. This model explains about 16% of variation in salary. However I have a lot of multicollinearity because my measures are all batting statistics and some measure similar things. This effects my standard errors in my model.

We could perform an F test to see how these metrics really effect salary and some more variables could be statistically significant in doing so.

Run regression using filters database

It is important to note the omitted category is Yankees

This regression model can show us if team impacts salary controlling for runs batting in (RBI) and batting average (#hits/#at bats)

```
model4 <- lm(log(Salary) ~ RBI + BattingAvg + Team, data = data1)
summary(model4)
```

```
##
## Call:
## lm(formula = log(Salary) ~ RBI + BattingAvg + Team, data = data1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0030 -0.8462 -0.1401  0.8725  3.2298
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   14.257204   0.506516  28.148 < 2e-16 ***
## RBI            0.015075   0.002506   6.015 4.66e-09 ***
## BattingAvg     0.361760   1.806012   0.200  0.8414
## TeamAngels    -0.187332   0.464477  -0.403  0.6870
## TeamAstros     0.096466   0.471829   0.204  0.8381
## TeamAthletics -1.000161   0.462939  -2.160  0.0314 *
## TeamBlue Jays -0.113536   0.471825  -0.241  0.8100
## TeamBraves     0.046916   0.454881   0.103  0.9179
## TeamBrewers    -0.624346   0.453926  -1.375  0.1699
## TeamCardinals  -0.474519   0.508443  -0.933  0.3513
## TeamCubs       -0.028513   0.461406  -0.062  0.9508
## TeamDiamondbacks -0.522881   0.470934  -1.110  0.2677
## TeamDodgers    0.905098   0.481134   1.881  0.0608 .
## TeamGiants     -0.547264   0.471346  -1.161  0.2464
## TeamGuardians  -0.648517   0.474502  -1.367  0.1726
## TeamMariners   0.001157   0.471208   0.002  0.9980
## TeamMarlins    -1.216316   0.463618  -2.624  0.0091 **
## TeamMets       0.280431   0.446847   0.628  0.5307
## TeamNationals  -0.735966   0.481561  -1.528  0.1274
## TeamOrioles    -0.157342   0.482107  -0.326  0.7444
## TeamPadres     0.012624   0.493818   0.026  0.9796
## TeamPhillies   0.152943   0.481286   0.318  0.7508
## TeamPirates    -0.416109   0.471057  -0.883  0.3777
## TeamRangers    -0.012721   0.471583  -0.027  0.9785
## TeamRays       -0.672576   0.463725  -1.450  0.1479
## TeamRed Sox    -0.147587   0.470638  -0.314  0.7540
## TeamReds       -0.508741   0.448214  -1.135  0.2572
## TeamRockies    -0.446410   0.455242  -0.981  0.3275
## TeamRoyals     -0.356271   0.472830  -0.753  0.4517
## TeamTigers     -0.776109   0.461986  -1.680  0.0939 .
## TeamTwins      -0.289255   0.471225  -0.614  0.5397
## TeamWhite Sox  -0.559602   0.457004  -1.225  0.2216
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.153 on 339 degrees of freedom
## Multiple R-squared:  0.2625, Adjusted R-squared:  0.1951
## F-statistic: 3.892 on 31 and 339 DF, p-value: 2.27e-10
```

The results of this regression show the same thing that RBI is statistically significant to salary but not batting average. This suggests when analyzing and determining salary, managers and owners care more about when you get hits and if you drive in runs rather than how many hits you get.

This regression also shows that some teams are statistically significant in determining your salary. This could be due to how much money the franchise has or by the location in where you play. The teams that show statistical significance are the Athletics and Marlins indicating that if you play for those teams you will make less compared to the New York Yankees (over 50 percent less).

The adjusted R squared is .195 so 19.5 percent of variation in salary is determined by this model.

Preferred Regression

Question being answered: Does team you play on affect batting performance? (In terms of hitting home runs)

I want to see if team and position affect OPS controlling for talent (WAR). War (wins above replacement) is calculated by combining offensive, defensive, and base running statistics and puts them in 1. I want to see if team or position affect OPS because baseball stadium dimensions are different at every ballpark along with the air and weather. I am using team as a proxy for home field.

```
model5 <-lm(HR ~ WAR + Team + Position, data = data2)
summary(model5)
```

```
##
## Call:
## lm(formula = HR ~ WAR + Team + Position, data = data2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.737  -4.521   0.000   4.040  23.479
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.0822     5.9722   2.860 0.005480 **
## WAR              3.2522     0.7597   4.281 5.44e-05 ***
## TeamAngels      2.4497     6.9242   0.354 0.724487
## TeamAstros     -6.8815     6.4099  -1.074 0.286452
## TeamAthletics  -3.8532     6.1668  -0.625 0.533978
## TeamBlue Jays  -9.6060     5.8669  -1.637 0.105751
## TeamBraves     -9.1925     7.1221  -1.291 0.200774
## TeamBrewers    -7.1492     6.0252  -1.187 0.239156
## TeamCardinals  -6.1805     7.2332  -0.854 0.395574
## TeamCubs       -6.5583     5.3651  -1.222 0.225382
## TeamDiamondbacks -5.8694     6.0244  -0.974 0.333050
## TeamDodgers    -6.9834     6.0751  -1.150 0.253992
## TeamGiants     -1.9303     6.4852  -0.298 0.766799
## TeamGuardians -18.1153     9.6599  -1.875 0.064641 .
## TeamMariners    4.4636     6.3391   0.704 0.483523
## TeamMarlins    -8.2799     6.9130  -1.198 0.234798
## TeamMets        1.5976     5.9783   0.267 0.790018
## TeamNationals   0.4689     7.7120   0.061 0.951678
## TeamOrioles    -12.0673     7.1247  -1.694 0.094467 .
## TeamPadres     -4.6625     6.9463  -0.671 0.504146
## TeamPhillies   -0.6754     6.1081  -0.111 0.912251
## TeamPirates    -5.2034    11.1619  -0.466 0.642443
## TeamRangers    -7.4825     6.3996  -1.169 0.246018
## TeamRays       -5.7288     6.4703  -0.885 0.378771
## TeamRed Sox    -6.7367     5.4692  -1.232 0.221896
## TeamReds       -4.7424     7.8070  -0.607 0.545387
## TeamRockies     1.0576     7.7917   0.136 0.892393
## TeamRoyals     -6.3519     7.8699  -0.807 0.422148
## TeamTigers     -2.0615     6.2995  -0.327 0.744387
## TeamTwins      -5.7827     6.7583  -0.856 0.394923
## Position1B      4.3391     3.5003   1.240 0.218981
## Position2B     -9.0370     3.2500  -2.781 0.006856 **
## Position3B     -4.4676     3.1145  -1.434 0.155604
## PositionC      -1.1921     3.2415  -0.368 0.714085
## PositionDH       7.3264     5.0203   1.459 0.148642
## PositionDH/RHP  19.8593     9.8670   2.013 0.047737 *
## PositionINF    -12.6841     6.0525  -2.096 0.039486 *
## PositionLHP    -20.7703     5.7649  -3.603 0.000563 ***
## PositionRHP    -21.9006     4.2250  -5.184 1.78e-06 ***
## PositionSS     -7.8676     2.7605  -2.850 0.005641 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 8.501 on 75 degrees of freedom
## Multiple R-squared:  0.6788, Adjusted R-squared:  0.5117
## F-statistic: 4.063 on 39 and 75 DF,  p-value: 9.645e-08
```

Interpreting Regression

This regression is asking if teams or position affect home runs controlling for player quality measured as WAR

The regression shows us that WAR (wins above replacement) is statistically significant to how many home runs players hit. For one additional point in WAR the home runs per season increase by 3.

The regression also shows that team is not statistically significant for hitting home runs per season. We don't have strong enough evidence to claim that Yankees hit more home runs because they play more games at Yankee stadium (which has one of the shortest right fields in baseball).

However the position players play do are significant in determining how many home runs players hit.

- Second baseman hit 9 home runs less than outfielders when controlling for team and player quality
- DH/RHP (which again is 1 person) hits 19 more home runs per season than outfielders when controlling for team and player quality
- Infielders hit 12 home runs less than outfielders when controlling for team and player quality
- Left handed pitchers hit 20 home runs less than outfielders when controlling for team and player quality
- Right handed pitchers hit 22 home runs less than outfielders when controlling for team and player
- Short Stop hit 8 home runs less than outfielders when controlling for team and player quality

Pitchers don't bat so the very high number makes sense as to why outfielders hit way more home runs per season than them.

The adjusted R square is also very high for this model. 51% of variation in home runs is explained by the model.

Shows that teams are mostly not statistically significant suggesting team affiliation does not strongly impact home runs hit.

Putting above regressions into 1 table

```
stargazer(model5, model4, model3,
  type = "html",
  title = "Regression Results",
  column.labels = c("How Positions Affect HR", "How Hits Affect Sal", "How Batting Metrics Affect Sal"),
  dep.var.labels = c("", "", ""),
  digits = 3,
  omit = "Team",
  dep.var.caption = "",
  add.lines = list(c("Team Fixed Effects", "Yes", "Yes", "No"))
)
```

Regression Results

How Positions Affect HR	How Hits Affect Sal	How Batting Metrics Affect Sal
(1)	(2)	(3)

WAR	3.252 ^{***} (0.760)		
HR			0.009 (0.016)
RBI		0.015 ^{***} (0.003)	0.012 ^{**} (0.006)
BattingAvg		0.362 (1.806)	0.377 (1.831)
Position1B	4.339 (3.500)		0.009 (0.227)
Position2B	-9.037 ^{***} (3.250)		-0.033 (0.221)
Position3B	-4.468 (3.115)		0.228 (0.231)
PositionC	-1.192 (3.242)		-0.067 (0.193)
PositionDH	7.326 (5.020)		0.430 (0.307)
PositionDH/RHP	19.859 ^{**} (9.867)		1.356 (1.243)
PositionINF	-12.684 ^{**} (6.053)		-0.562 ^{**} (0.257)
PositionLHP	-20.770 ^{***} (5.765)		
PositionRHP	-21.901 ^{***} (4.225)		-0.578 (1.185)
PositionSS	-7.868 ^{***} (2.760)		0.151 (0.218)
Constant	17.082 ^{***} (5.972)	14.257 ^{***} (0.507)	13.953 ^{***} (0.384)
Team Fixed Effects	Yes	Yes	No
Observations	115	371	371
R ²	0.679	0.263	0.187
Adjusted R ²	0.512	0.195	0.160
Residual Std. Error	8.501 (df = 75)	1.153 (df = 339)	1.178 (df = 358)
F Statistic	4.063 ^{***} (df = 39; 75)	3.892 ^{***} (df = 31; 339)	6.858 ^{***} (df = 12; 358)

Note:

 $p < 0.1$; $p < 0.05$; $p < 0.01$

Question being answered: Do traditional or advanced statistics affect salary more?

```

model6 <- lm(log(Salary) ~ HR + BattingAvg + RBI + Position, data = data1) # traditional stats only

model7 <- lm(log(Salary) ~ WAR + OPS + Position, data = data2) #advanced stats only

model8 <- lm(log(Salary) ~ HR + BattingAvg + RBI + WAR + Position, data = data2) # advanced plus traditional stats

stargazer(model6, model7, model8,
  type = "html",
  title = "Regression Results",
  column.labels = c("Trad Stats on Sal", "Advance Stats on Sal", "All on Sal"),
  dep.var.labels = c("", "", ""),
  digits = 3,
  omit = "Position",
  dep.var.caption = "",
  add.lines = list(c("Position Fixed Effects", "Yes", "Yes", "Yes")))
)

```

Regression Results

	Trad Stats on Sal (1)	Advance Stats on Sal (2)	All on Sal (3)
HR	0.009 (0.016)		0.014 (0.027)
BattingAvg	0.377 (1.831)		-2.075 (1.526)
RBI	0.012** (0.006)		0.009 (0.012)
WAR		0.190* (0.112)	0.093 (0.121)
OPS		-0.430 (0.764)	
Constant	13.953*** (0.384)	14.890*** (0.698)	14.393*** (0.742)
Position Fixed Effects	Yes	Yes	Yes
Observations	371	115	115
R ²	0.187	0.080	0.129
Adjusted R ²	0.160	-0.029	0.007
Residual Std. Error	1.178 (df = 358)	1.439 (df = 102)	1.414 (df = 100)
F Statistic	6.858*** (df = 12; 358)	0.735 (df = 12; 102)	1.053 (df = 14; 100)

Note:

$p < 0.1$; **$p < 0.05$** ; $p < 0.01$

Results explained

This table compares explanatory power of traditional and advanced performance metrics in determining MLB players salary

The traditional statistic model explains about 16% of variation in log salary

The advanced statistic model performs worse having a negative adjusted R squared. It does show that WAR is marginally significant and OPS is not statistically significant most likely due to the overlap with WAR. The coefficient on OPS doesn't indicate a negative relationship with salary, however it indicates that OPS does not provide any additional information when WAR is controlled for.

In the combined model explanatory power declines as multicollinearity increases.

Overall, the results show that salary is not strongly determined by either traditional or advanced offensive statistics alone, and much of the variation in salary is explained by other factors not included in my models.

These conclusions might be skewed because of multicollinearity and should be analyzed further with F tests and other tools.